

Theorem 1. Let $\mathbf{x} := (\mathbf{y}, \mathbf{z})$ and define

$$\mathbf{F}(\mathbf{x}) := \begin{pmatrix} \mathbf{G}(\mathbf{y}, \mathbf{z}) \\ \mathbf{H}(\mathbf{y}, \mathbf{z}) \end{pmatrix}, \quad \mathbf{F} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^{n+m}.$$

Let $(\mathbf{y}_0, \mathbf{z}_0)$ be given.

Assume there exists an open neighbourhood $\mathcal{U} \subset \mathbb{R}^{n+m}$ of a solution $\mathbf{x}^* = (\mathbf{y}^*, \mathbf{z}^*)$ such that:

1. $\mathbf{G}, \mathbf{H} \in C^1(\mathcal{U})$;
2. the partial Jacobians

$$\partial_{\mathbf{y}} \mathbf{G}(\mathbf{y}, \mathbf{z}), \quad \partial_{\mathbf{z}} \mathbf{H}(\mathbf{y}, \mathbf{z})$$

are uniformly nonsingular on $\mathcal{U}$;

3. there exist constants $\delta_0^{(z)}, \delta_0^{(y)} > 0$ and $\delta_{\max}^{(z)}, \delta_{\max}^{(y)} < \infty$ such that the adaptive pseudo-time steps defined below satisfy

$$0 < \delta_{k,l}^{(z)} \leq \delta_{\max}^{(z)}, \quad 0 < \delta_k^{(y)} \leq \delta_{\max}^{(y)};$$

4. all inner and outer iterates generated by the algorithm remain in $\mathcal{U}$.

For each $k \geq 0$, define the inner pseudo-transient continuation iteration $\{\mathbf{z}_k^l\}_{l \geq 0}$ by

$$\left((\delta_{k,l}^{(z)})^{-1} \mathbf{I} + \partial_{\mathbf{z}} \mathbf{H}(\mathbf{y}_k, \mathbf{z}_k^l) \right) (\mathbf{z}_k^{l+1} - \mathbf{z}_k^l) = -\mathbf{H}(\mathbf{y}_k, \mathbf{z}_k^l),$$

with adaptive time-step

$$\delta_{k,l}^{(z)} = \min \left(\delta_0^{(z)} \frac{\|\mathbf{H}(\mathbf{y}_k, \mathbf{z}_k^0)\|}{\|\mathbf{H}(\mathbf{y}_k, \mathbf{z}_k^l)\|}, \delta_{\max}^{(z)} \right).$$

Define

$$\mathbf{z}_k^* := \lim_{l \rightarrow \infty} \mathbf{z}_k^l,$$

whose existence is guaranteed by pseudo-transient continuation theory.

The outer iteration for $\mathbf{y}$ is defined by

$$\left((\delta_k^{(y)})^{-1} \mathbf{I} + \partial_{\mathbf{y}} \mathbf{G}(\mathbf{y}_k, \mathbf{z}_k^*) \right) (\mathbf{y}_{k+1} - \mathbf{y}_k) = -\mathbf{G}(\mathbf{y}_k, \mathbf{z}_k^*),$$

with adaptive time-step

$$\delta_k^{(y)} = \min \left(\delta_0^{(y)} \frac{\|\mathbf{F}(\mathbf{x}_0)\|}{\|\mathbf{F}(\mathbf{y}_k, \mathbf{z}_k^*)\|}, \delta_{\max}^{(y)} \right).$$

Then the sequence $\{(\mathbf{y}_k, \mathbf{z}_k^*)\}$ converges to $\mathbf{x}^* = (\mathbf{y}^*, \mathbf{z}^*)$, and

$$\mathbf{F}(\mathbf{x}^*) = \mathbf{0}.$$

Proof. Fix k and consider the inner problem for $\mathbf{z}$ with $\mathbf{y}_k$ held fixed. Define

$$\mathbf{H}_{\mathbf{y}_k}(\mathbf{z}) := \mathbf{H}(\mathbf{y}_k, \mathbf{z}).$$

By Assumption (1), $\mathbf{H}_{\mathbf{y}_k} \in C^1$, and by Assumption (2), $\partial_{\mathbf{z}} \mathbf{H}(\mathbf{y}_k, \mathbf{z})$ is nonsingular in $\mathcal{U}$.

The inner iteration is precisely the pseudo-transient continuation method applied to the nonlinear equation

$$\mathbf{H}_{\mathbf{y}_k}(\mathbf{z}) = \mathbf{0},$$

with adaptive pseudo-time step $\delta_{k,l}^{(z)}$ that is uniformly bounded above and bounded away from zero by Assumption (3). Therefore, by the global convergence theorem for pseudo-transient continuation [1], the sequence $\{\mathbf{z}_k^l\}_{l \geq 0}$ converges to a limit

$$\mathbf{z}_k^* := \lim_{l \rightarrow \infty} \mathbf{z}_k^l,$$

satisfying

$$\mathbf{H}(\mathbf{y}_k, \mathbf{z}_k^*) = \mathbf{0}.$$

Thus, for each outer iteration index k , the inner pseudo-transient continuation defines a reduced state $\mathbf{z}_k^*$.

The Implicit Function Theorem guarantees that the reduced state $\mathbf{z}_k^*$ depends smoothly on $\mathbf{y}_k$ in a neighbourhood of $\mathbf{y}^*$, and that the solution of $\mathbf{H}(\mathbf{y}, \mathbf{z}) = \mathbf{0}$ is locally unique. Consequently, the reduced outer problem obtained by fixing $\mathbf{z} = \mathbf{z}_k^*$ is well defined.

Next, define the reduced outer operator

$$\mathbf{G}_{\mathbf{z}_k^*}(\mathbf{y}) := \mathbf{G}(\mathbf{y}, \mathbf{z}_k^*).$$

By Assumption (1), $\mathbf{G}_{\mathbf{z}_k^*} \in C^1$, and by Assumption (2), $\partial_{\mathbf{y}} \mathbf{G}(\mathbf{y}, \mathbf{z}_k^*)$ is nonsingular in a neighbourhood of $\mathbf{y}^*$.

The outer iteration is again a pseudo-transient continuation method, now applied to the reduced equation

$$\mathbf{G}_{\mathbf{z}_k^*}(\mathbf{y}) = \mathbf{0},$$

with adaptive pseudo-time step $\delta_k^{(y)}$ satisfying the same boundedness properties.

Applying the pseudo-transient convergence theorem once more yields

$$\mathbf{y}_k \xrightarrow[k \rightarrow \infty]{} \mathbf{y}^* \quad \text{and} \quad \mathbf{G}(\mathbf{y}^*, \mathbf{z}^*) = \mathbf{0}.$$

Finally, since $\mathbf{H}(\mathbf{y}_k, \mathbf{z}_k^*) = \mathbf{0}$ for all k and $\mathbf{H}$ is continuous by Assumption (1), passing to the limit gives

$$\mathbf{H}(\mathbf{y}^*, \mathbf{z}^*) = \mathbf{0}.$$

Therefore,

$$\mathbf{F}(\mathbf{x}^*) = \mathbf{0},$$

which completes the proof. □

References

- [1] C. T. Kelley, D. E. Keyes, Convergence analysis of pseudo-transient continuation, SIAM Journal on Numerical Analysis 35 (2) (1998) 508–523.